Chapter 1
The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

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1.5 Nested Quantifiers

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Predicate

【**Definition**】A predicate (propositional function) is a statement that contains variables. Once the values of the variables are specified, the function has a truth value.

- $P(x_1, x_2, \cdots, x_n)$: n-place (n-ary) predicate
- Examples
  - $P(x)$="$x>3$"
  - $Q(x,y)$="x is the best player of team y"
  - $R(x,y,z)$="$x+y=z$"

Predicate

- Examples
  - Let $P(x)$ denote the statement "$x > 0$." What are the truth values of $P(-3)$, $P(0)$ and $P(3)$?

    $P(-3)$="-3>0" F

    $P0)$="0>0" F

    $P(3)$="3>0" T

  - Let $Q(x, y)$ denote the statement "$x < y$." What are the truth values of $Q(4, 3)$ and $Q(2, 7)$?

    $Q(4, 3)$ ="4 < 3" F

    $Q(2, 7)$ = "2 < 7" T

Quantifiers

- Quantifications
  - Every computer connected to the university network is functioning properly.
  - Students in this class are smart.
- Universal quantification
- Existential quantification
- Predicate logic (calculus): the area of logic that deals with predicate and quantifiers.

Universal Quantification

【**Definition**】A universal quantification of $P(x)$, denoted by $\forall\, x\, P(x)$, is the statement "$P(x)$ for all values of x in the domain. "

- $\forall$: universal quantifier
- Domain (domain of discourse / universe of discourse): range of the possible values of the variable x
- An element for which $P(x)$ is false is called a counterexample of $\forall\, x\, P(x)$
- Example:

  Let $P(x)$ be the statement "$x>0$." In the domain of all integers, x= -1 is a counterexample for $\forall\, x\, P(x)$.

Universal Quantification

- Many ways to express universal quantification:
  - For all
  - For every
  - All of
  - For each
  - Given any
  - For arbitrary
  - For any

Universal Quantification

- Examples:

1. What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

Solution:

(1) If the domain consists of all real numbers.

Because $\left(\frac{1}{2}\right)^2 < \frac{1}{2}$, $\forall x(x^2 \geq x)$ is false

(2) If the domain consists of all integers.

$\forall x(x^2 \geq x)$ is true

Universal Quantification

- Examples:

2. Express the following statement as a universal quantification: "All lions are fierce."

Solution:

Let $Q(x)$ denote the statement "x is fierce".

(1) Assuming that the domain is the set of all lions.

$$\forall x Q(x)$$

(2) Assuming that domain is the set of all creatures.

Let $P(x)$ denote the statement "x is a lion".

$$\forall x(P(x) \rightarrow Q(x))$$

Universal Quantification

- Examples:

3. What is the truth value of $\forall x\, P(x)$, where $P(x)$ is the statement "$x<3$" and the domain is $\{1,2,3\}$?

Solution:

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$$

Because $P(3)$, which is the statement "$3<3$," is false, it follows that $\forall x\, P(x)$ is false.

- Remark: Given the domain as $\{x_1, x_2, \cdots, x_n\}$,

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n)$$

Existential Quantification

【**Definition**】 An existential quantification of $P(x)$, denoted by $\exists x\, P(x)$, is the statement "There exists an element x in the domain such that $P(x)$. "

- $\exists$: existential quantifier
- Other expressions:

  For some x $P(x)$

  There is an x such that $P(x)$

  There is at least one x such that $P(x)$

Existential Quantification

- Examples:

1. Express the following statement as an existential quantification.

   "Some real numbers are rational numbers. "

Solution:

Let $Q(y)$: y is a rational number

(1) Assuming that the domain is the set of all real numbers.

$$\exists y Q(y)$$

(2) Assuming that the domain is the set of all complex numbers. Let $R(y)$: y is a real number

$$\exists y(R(y) \wedge Q(y))$$

Existential Quantification

- Examples:

2. What is the truth value of $\exists x\, P(x)$, where $P(x)$ is the statement "$x<3$" and the domain is {1,2,3} ?

Solution:

$$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$$

Because $P(1)$, which is the statement "1<3," is true, it follows that $\exists x\, P(x)$ is true.

- Remark: Given the domain as $\{x_1, x_2, \cdots, x_n\}$,

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n)$$

Quantifiers

Statement	When true?	When false?
$\forall x\, P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x\, P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

- Other Quantifiers
  - Uniqueness quantifier: $\exists!$ or $\exists_1$

    $\exists!\, P(x)$ or $\exists_1 P(x)$: There exists a unique x such that $P(x)$ is true.

Quantifiers with Restricted Domains

- Example: What do the statements $\forall x<0\ (x^2>0)$, $\exists y>0\ (y^2=2)$ mean, where the domain in each case consists of the real numbers?

Solution:

$$\forall x<0\ (x^2>0)$$

For every real number x with $x<0$, $x^2>0$

$$\exists y>0\ (y^2=2)$$

There exists a real number y with $y>0$ such that $y^2=2$

Precedence of Quantifiers

- The quantifiers ∃ and ∀ have higher precedence than all logical operators from propositional calculus.
- Example:

$$\forall x P(x) \vee Q(x)$$

$$(\forall x P(x)) \vee Q(x) \quad \surd$$

$$\forall x (P(x) \vee Q(x)) \quad \mathbf{X}$$

Binding Variables

- Bound variable: a variable is bound if it is quantified.
- Free variable: a variable neither quantified nor specified with a value
- *All the variables in a propositional function must be quantified or set equal to a particular value to turn it into a proposition.*
- Scope of a quantifier: the part of a logical expression to which the quantifier is applied
- Examples

$\exists x\ (x+y)=1$

$\exists x\ (P(x) \wedge Q(x)) \vee \forall x\ R(x)$

Logical Equivalences Involving Quantifiers

【Definition】 Statements involving predicates and quantifiers are *logically equivalent* iff they have the same truth

- *for every predicate substituted into these statements and*
- *for every domain of discourse used for the variables in the expressions.*

- $S \equiv T$: two statements S and T involving predicates and quantifiers are logically equivalent.
- Examples (Assuming all variables are in the same domain):

$$\forall x(A(x) \wedge B(x)) \quad \equiv \quad \forall x A(x) \wedge \forall x B(x)$$

$$\exists x(A(x) \vee B(x)) \quad \equiv \quad \exists x A(x) \vee \exists x B(x)$$

$$\forall x(A(x) \vee B(x)) \quad \not\equiv \quad \forall x A(x) \vee \forall x B(x)$$

$$\exists x(A(x) \wedge B(x)) \quad \not\equiv \quad \exists x A(x) \wedge \exists x B(x)$$

Negating Quantified Expressions

- De Morgan's laws for quantifiers

$$\neg\forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg\exists x P(x) \equiv \forall x \neg P(x)$$

- Example

Let $P(x)$ be the statement "Student x has taken calculus." Assume the domain consists of the students in our class.

1. Express $\forall x\, P(x)$ and its negation.
2. Express $\exists x\, P(x)$ and its negation.

Solution:

$\forall x\, P(x)$: Every student in our class has taken calculus.

$\neg\forall x P(x)$: There is a student in our class who has not taken calculus. ($\exists x \neg P(x)$)

$\exists\, x\, P(x)$: There is a student in our class who has taken calculus.

$\neg\exists x P(x)$: Every student in our class has not taken calculus. ($\forall x \neg P(x)$)

De Morgan's Laws for Quantifiers

Negation	Equivalent Statement	When is Negation True?	When False?
$\neg\exists x\, P(x)$	$\forall x\, \neg P(x)$	$P(x)$ is false for every x	There is an x for which $P(x)$ is true
$\neg\forall x\, P(x)$	$\exists x\, \neg P(x)$	There is an x for which $P(x)$ is false	$P(x)$ is true for every x

Logical Equivalences Involving Quantifiers

Proof:

$$\forall x(P(x) \rightarrow A) \equiv \forall x(\neg P(x) \vee A)$$

$$\equiv \forall x \neg P(x) \vee A$$

$$\equiv \neg \exists x P(x) \vee A$$

$$\equiv \exists x P(x) \rightarrow A$$

Proof:

$$\forall x(A \rightarrow P(x)) \equiv \forall x(\neg A \vee P(x))$$

$$\equiv \neg A \vee \forall x P(x)$$

$$\equiv A \rightarrow \forall x P(x)$$

Translating from English into Logical Expressions

- **Goal:** To produce a logical expression that is simple and can be easily used in subsequent reasoning.
- **Steps:**
  - Clearly identify the appropriate quantifier(s)
  - Introduce variable(s) and predicate(s)
  - Translate using quantifiers, predicates, and logical operators

There can be many ways to translate a particular sentence.

Example

- $C(x)$: x is a CS student, $E(x)$: x is a Math student, $S(x)$: x is a smart student, and the domain consists of all students in our class

1) Everyone is a CS student.

$$\forall x\, C(x)$$

2) Nobody is a Math student.

$$\forall x \neg E(x) \quad \textbf{or} \quad \neg \exists x\, E(x)$$

3) All CS students are smart students.

$$\forall x\, (C(x) \rightarrow S(x))$$

4) Some CS students are smart students.

$$\exists x\, (C(x) \wedge S(x))$$

Example

- $C(x)$: x is a CS student, $E(x)$: x is an Math student, $S(x)$: x is a smart student, and the domain consists of all students in our class

5) No CS student is an Math student.

- If x is a CS student, then that student is not a Math student.

$$\forall x\,(C(x) \rightarrow \neg\, E(x))$$

- There does not exist a CS student who is also a Math student.

$$\neg\, \exists x\,[C(x) \wedge E(x)]$$

6) If any Math student is a smart student then he is also a CS student.

$$\forall x\,((E(x) \wedge S(x)) \rightarrow C(x))$$

Examples from Lewis Carroll [1]

- "All lions are fierce"

  "Some lions do not drink coffee"

  "Some fierce creatures do not drink coffee"

Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a lion", "x is fierce", and "x drinks coffee", respectively. Assume the domain consists of all creatures.

$$\forall x\,(P(x) \rightarrow Q(x))$$

$$\exists \mathrm{x}\,(P(x) \wedge \neg\, R(x))$$

$$\exists \mathrm{x}\,(Q(x) \wedge \neg\, R(x))$$

Examples from Lewis Carroll [2]

- "All hummingbirds are richly colored"

  "No large birds live on honey"

  "Birds that do not live on honey are dull in color"

  "Hummingbirds are small"

Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements "x is a hummingbird", "x is large", "x lives on honey", and "x is richly colored", respectively. Assume the domain consists of all birds.

$$\forall x\,(P(x) \rightarrow S(x))$$

$$\neg\exists \mathrm{x}\,(Q(x) \wedge R(x))$$

$$\forall \mathrm{x}\,(\neg R(x) \rightarrow \neg S(x))$$

$$\forall \mathrm{x}\,(P(x) \rightarrow \neg Q(x))$$

Translating from English into Logical Expressions

- **Tips**
  - All ***S*(*x*)** are ***O*(*x*)** : $\forall x\,(S(x) \rightarrow O(x))$
  - No ***S*(*x*)** are ***O*(*x*)** : $\forall x\,(S(x) \rightarrow \neg\, O(x))$
  - Some ***S*(*x*)**'s are ***O*(*x*)** : $\exists x\,(S(x) \wedge O(x))$
  - Some ***S*(*x*)** are not ***O*(*x*)** : $\exists x\,(S(x) \wedge \neg\, O(x))$

Where ***S*(*x*)** and ***O*(*x*)** are propositional functions of variable ***x***

Homework

Due on Mar. 14 (Tuesday)

Ver. 7

Sec. 1.4

6(c,d,e,f), 9(b,d), 20(e), 24(b,d), 40(b), 44, 49(a), 60

Ver. 8

Sec. 1.4

6(c,d,e,f), 9(b,d), 20(e), 24(b,d), 42(b), 46, 51(a), 62

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Nested Quantifiers

- **A statement can have multiple variables**
- Example:

  Let $C(x,y)$ denote "x has taken course y," where the domain of x and y consist of all ZJU students and all CS courses, respectively. We may construct the following propositions:

  $$\forall x \exists y C(x, y)$$

  Every student at ZJU has taken some CS course.

  $$\exists x \forall y C(x, y)$$

  Some student at ZJU has taken all CS courses.

【**Definition**】Two quantifiers are nested if one is within the scope of the other.

Translating from Nested Quantifiers into English

- Examples

1. Translate the statement $\forall x(C(x) \vee \exists y(C(y) \wedge F(x,y)))$ into English, where $C(x)$ is "x has a computer," $F(x,y)$ is "x and y are friends," and the domain for both x and y consists of all students at ZJU.

Solution:

For every student x at ZJU, x has a computer or there is a student y such that y has a computer and x and y are friends.



Every student at ZJU has a computer or has a friend who has a computer.

Translating from Nested Quantifiers into English

- Examples

2. Translate the following statement into English, where $F(x,y)$ means x and y are friends and the domain for x, y, and z consists of all students at ZJU.

$$\exists x \forall y \forall z(((F(x,y) \wedge F(x,z) \wedge (y \neq z)) \rightarrow \neg F(y,z)))$$

Solution:

There is a student x such that for all students y and all students z other than y, if x and y are friends and x and z are friends, then y and z are not friends.

⇳

There is a student none of whose friends are also friends with each other.

Translating English into Logical Expressions

- Examples

1. Express the statement "Everyone has exactly one best friend" as a logical expression with a domain consisting of all people.

Solution:

Rewrite the original statement as

"For every person x, x has exactly one best friend."

⇩

"There is a person y who is the best friend of x, and furthermore, that for every person z, if z is not y, then z is not the best friend of x."

Let $B(x, y)$ be the statement "y is the best friend of x."

$$\forall x \exists y \forall z (B(x, y) \wedge ((z \neq y) \rightarrow \neg B(x, z)))$$

Translating English into Logical Expressions

- Examples

2. Express the statement "If a person is female and is a parent, then this person is someone's mother" as a logical expression with a domain consisting of all people.

Solution:

Rewrite the original statement as

"For every person x, if x is female and x is a parent, then there exists a person y such that x is the mother of y."

Let $F(x)$: "x is female,"

$P(x)$: "x is a parent,"

$M(x, y)$: "x is the mother of y."

$$\forall x((F(x) \wedge P(x)) \rightarrow \exists y M(x, y))$$

$$\forall x \exists y((F(x) \wedge P(x)) \rightarrow M(x, y))$$

Translating English into Logical Expressions

- Examples

3. Express the definition of a limit using quantifiers.

Solution:

The definition of the limit $\lim_{x \to a} f(x) = L$ is:

"For every real number $\varepsilon > 0$ there exists a real number $\delta > 0$ such that $|f(x) - L| < \varepsilon$ *wherever* $|x - a| < \delta$ "

Assume the domain for the variables ε, δ, and x consist of all real numbers, we have

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x (0 < |x - a| < \delta \rightarrow |f(x) - L| < \varepsilon)$$

The Order of Quantifiers

- **The order of nested quantifiers matters if quantifiers are of different types**

$\forall x \forall y P(x, y) \equiv \forall y \forall x P(x, y)$

$\exists x \forall y P(x, y)$ **is not the same as** $\forall y \exists x P(x, y)$

- Example:

Assume P(x,y) denote "x loves y," where the domain for variables x and y consists of all people. Translate the following expressions into English

$\exists x \forall y P(x, y)$

There is someone who loves everyone.

$\forall y \exists x P(x, y)$

Everybody is loved by somebody.

Quantifications of Two Variables

Statement	*When True?*	*When False?*
$\forall x \, \forall y \, P(x, y)$ $\forall y \, \forall x \, P(x, y)$	$P(x, y)$ is true for every pair x, y.	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \, \exists y \, P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x for which $P(x, y)$ is false for every y.
$\exists x \, \forall y \, P(x, y)$	There is an x for which $P(x, y)$ is true for every y.	For every x there is a y for which $P(x, y)$ is false.
$\exists x \, \exists y \, P(x, y)$ $\exists y \, \exists x \, P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y.

Negating Nested Quantifiers

- **Negating nested quantifiers by successively applying the rules for negating statements involving a single quantifier**
- **Examples:**

1. Express the negation of the statement $\forall x \exists y\ (xy=1)$ so that no negation precedes a quantifiers.

Solution:

$$\neg(\forall x \exists y\ (xy=1))$$

$$\equiv \exists x(\neg \exists y\ (xy=1))$$

$$\equiv \exists x \forall y\ (\neg\ (xy=1))$$

$$\equiv \exists x \forall y\ (xy \neq 1)$$

Negating Nested Quantifiers

- **Negating nested quantifiers by successively applying the rules for negating statements involving a single quantifier**
- **Examples:**

2. Use quantifiers and predicates express the fact that $\lim_{x \to a} f(x)$ does not exist.

Solution:

$$\neg(\forall \varepsilon > 0 \exists \delta > 0 \forall x(0 < |x-a| < \delta \rightarrow |f(x)-L| < \varepsilon))$$

$$\equiv \exists \varepsilon > 0 \neg \exists \delta > 0 \forall x(0 < |x-a| < \delta \rightarrow |f(x)-L| < \varepsilon)$$

$$\equiv \exists \varepsilon > 0 \forall \delta > 0 \neg \forall x(0 < |x-a| < \delta \rightarrow |f(x)-L| < \varepsilon)$$

$$\equiv \exists \varepsilon > 0 \forall \delta > 0 \exists x \neg(0 < |x-a| < \delta \rightarrow |f(x)-L| < \varepsilon)$$

$$\equiv \exists \varepsilon > 0 \forall \delta > 0 \exists x(0 < |x-a| < \delta \wedge |f(x)-L| \geq \varepsilon)$$

$$\forall L \exists \varepsilon > 0 \forall \delta > 0 \exists x(0 < |x-a| < \delta \wedge |f(x)-L| \geq \varepsilon)$$

Homework

Due on Mar. 14 (Tuesday)

Sec. 1.5

Ver. 7 & Ver. 8

6(e, f), 12(d, h, k, n), 14(c, d, e, f), 24(a, d), 32(d), 34, 38(b, d), 42

Functionally Complete

【Definition】 A set of logical operators is called ***functionally complete*** if every compound proposition is logically equivalent to a compound proposition involving only this set of logical operators.

- $\wedge$, $\vee$, and $\neg$ form a functionally complete set of operators.
- $\wedge$ ($\vee$) and $\neg$ form a functionally complete set of operators.

Propositional Normal Forms

- Literal: ***p* or ¬*p***

【**Definition**】 Disjunctions (conjunctions) with one or more literals as disjuncts (conjuncts) are called ***disjunctive (conjunctive) clauses*****.** Disjunctive and conjunctive clauses are simply called clauses.

- Examples:
  - $q \vee r$ Disjunctive clause
  - $\neg q \wedge p$ Conjunctive clause
  - $p \wedge q \vee r$ Not a clause

Conjunctive Normal Form (CNF)

【**Definition**】A conjunction with one or more disjunctive clauses as its conjuncts is said to be in ***conjunctive normal form***.

$$(A_{11} \vee \ldots \vee A_{1n_1}) \wedge \ldots \wedge (A_{k1} \vee \ldots \vee A_{kn_K})$$

- Examples:
  - $p \wedge (q \vee r)$ √
  - $\neg q \wedge p$ √
  - $p \wedge ((p \wedge q) \vee r)$ x

Disjunctive Normal Form (DNF)

【**Definition**】A disjunction with one or more conjunctive clauses as its disjuncts is said to be in ***disjunctive normal form***.

$$(A_{11} \wedge \ldots \wedge A_{1n_1}) \vee \ldots \vee (A_{k1} \wedge \ldots \wedge A_{kn_K})$$

- Examples:
  - $(p \wedge q) \vee (p \wedge \neg q)$ √
  - $p \vee (q \wedge r)$ √
  - $\neg p \vee \mathrm{T}$ √
  - $\neg(p \wedge q) \vee r$ x